

Interactive "What-If" Scenario Simulator

A straightforward, non-technical manual for stress-testing pricing, client churn, break-even thresholds, and cash runway for free.

Zero-SaaS, 100% Client-Side Business Intelligence

Static spreadsheets break the moment you tweak a formula, and enterprise forecasting software requires expensive monthly subscriptions. This tool lives in a single, lightweight HTML file on your laptop. You double-click it, drag the sliders, and immediately see how pricing adjustments, churn, and operational overhead impact your 12-month net profit curve in real time.

1. Overview: How It Works

There are no complex installations, databases, or cloud servers. The entire simulator operates locally in any modern browser (Chrome, Edge, Safari, Firefox):

The Controls (Inputs)

Sliders for Retainer/Price, Starting Clients, New Sales/Month, Churn Rate %, Fixed OpEx, and Direct Costs (COGS).

Real-Time KPIs

Instantly computes your Break-Even Client Count, Net Unit Margin (\$/client), and Month 1 vs. Month 12 Net Profits.

Dynamic Trajectory Chart

Draws a 12-month visual curve comparing Gross Revenue, Total Expenses, and Net Monthly Profit.

100% Privacy & Zero Cost

No data ever leaves your computer. No cookies, no telemetry, and zero subscription charges.

2. How to Set Up in 30 Seconds

STEP 1

Create the File

Open **Notepad** on your computer, paste the complete code from Section 3, and save it as `calculator.html`.

STEP 2

Open in Your Browser

Double-click `calculator.html`. It will instantly launch in your browser with interactive sliders and real-time Chart.js graphics.

3. The Complete Source Code (calculator.html)

Copy and paste this entire code into a blank text document and name it `calculator.html` :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>What-If Scenario Simulator | 3Sci Open Source Work Hack</title>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <link href="https://fonts.googleapis.com/css2?family=PlusJakarta+Sans:wght@400;600;700;800&family=Space+Grotesk:wght@700&display=swap" rel="stylesheet">
  <style>
    :root { --bg: #090f17; --card-bg: #111a26; --card-border: #1e2c3d; --accent: #00d2b4; --accent-dim: rgba(0,210,180,0.15); --text: #f1f5f9; --text-muted: #94a3b8; --profit: #10b981; --costs: #ef4444; --rev: #38bdf8; }
    * { box-sizing: border-box; margin: 0; padding: 0; font-family: 'Plus Jakarta Sans', sans-serif; }
    body { background-color: var(--bg); color: var(--text); padding: 30px 40px; min-height: 100vh; }
    header { display: flex; justify-content: space-between; align-items: center; border-bottom: 2px solid var(--card-border); padding-bottom: 16px; margin-bottom: 24px; }
    .brand-group { display: flex; align-items: center; gap: 16px; }
    .brand-logo { height: 48px; width: auto; object-fit: contain; }
    .brand-title { font-family: 'Space Grotesk', sans-serif; font-size: 22px; font-weight: 700; color: var(--accent); }
    .badge { background: var(--accent-dim); color: var(--accent); border: 1px solid var(--accent); padding: 6px 14px; border-radius: 6px; font-size: 13px; font-weight: 700; }
    .grid-container { display: grid; grid-template-columns: 420px 1fr; gap: 24px; }
    .panel { background: var(--card-bg); border: 1px solid var(--card-border); border-radius: 12px; padding: 24px; }
    .panel-header { display: flex; justify-content: space-between; align-items: center; margin-bottom: 20px; border-bottom: 1px solid var(--card-border); padding-bottom: 10px; }
    .presets { display: flex; gap: 6px; }
    .btn-preset { background: #1c2b3d; color: #cbd5e1; border: 1px solid #2d3f55; padding: 4px 8px; font-size: 11px; font-weight: 700; border-radius: 4px; cursor: pointer; }
    .control-group { margin-bottom: 18px; }
    .control-label { display: flex; justify-content: space-between; font-size: 13px; font-weight: 700; color: var(--text-muted); margin-bottom: 8px; }
    .control-val { color: #fff; font-size: 15px; font-weight: 800; }
    input[type=range] { width: 100%; height: 6px; background: #233448; border-radius: 3px; outline: none; -webkit-appearance: none; }
    input[type=range]::-webkit-slider-thumb { -webkit-appearance: none; width: 18px; height: 18px; border-radius: 50%; background: var(--accent); cursor: pointer; }
    .kpi-row { display: grid; grid-template-columns: repeat(4, 1fr); gap: 16px; margin-bottom: 24px; }
    .kpi-card { background: var(--card-bg); border: 1px solid var(--card-border); border-radius: 10px; padding: 16px 20px; }
    .kpi-label { font-size: 12px; font-weight: 700; text-transform: uppercase; color: var(--text-muted); }
    .kpi-val { font-size: 26px; font-weight: 800; margin: 8px 0 4px 0; font-family: 'Space Grotesk', sans-serif; }
    .kpi-val.good { color: var(--profit); } .kpi-val.rev { color: var(--rev); } .kpi-val.alert { color: var(--costs); }
    .chart-box { background: var(--card-bg); border: 1px solid var(--card-border); border-radius: 12px; padding: 24px; height: 480px; display: flex; flex-direction: column; }
    footer { display: flex; justify-content: space-between; align-items: center; border-top: 1px solid var(--card-border); margin-top: 28px; padding-top: 14px; font-size: 13px; color: var(--text-muted); }
    .footer-link { color: var(--accent); text-decoration: none; font-weight: 700; }
  </style>
</head>
<body>
  <header>
    <div class="brand-group">
      
      <div>
        <div class="brand-title">3SCI SCENARIO SIMULATOR</div>
        <div style="font-size: 13px; color: var(--text-muted);">Interactive What-If Cash Flow, Break-Even & Unit Economics Engine</div>
      </div>
    </div>
    <div class="badge">Open Source Hack #2</div>
  </header>

  <div class="grid-container">
    <div class="panel">
      <div class="panel-header">
```

```

    <span style="font-weight:800; color:var(--accent);">PARAMETERS</span>
    <div class="presets">
      <button class="btn-preset" onclick="setPreset('conservative')">Cautious</button>
      <button class="btn-preset" onclick="setPreset('base')">Base</button>
      <button class="btn-preset" onclick="setPreset('aggressive')">Growth</button>
    </div>
  </div>
  <div class="control-group">
    <div class="control-label"><span>Monthly Price</span><span class="control-val" id="val-price">$250</span></div>
    <input type="range" id="param-price" min="50" max="1500" step="25" value="250" oninput="updateModel()"/>
  </div>
  <div class="control-group">
    <div class="control-label"><span>Active Clients</span><span class="control-val" id="val-clients">30</span></div>
    <input type="range" id="param-clients" min="1" max="150" step="1" value="30" oninput="updateModel()"/>
  </div>
  <div class="control-group">
    <div class="control-label"><span>New Clients / Mo</span><span class="control-val" id="val-new-clients">5</span></div>
    <input type="range" id="param-new-clients" min="0" max="30" step="1" value="5" oninput="updateModel()"/>
  </div>
  <div class="control-group">
    <div class="control-label"><span>Monthly Churn Rate</span><span class="control-val" id="val-churn">5.0%</span></div>
    <input type="range" id="param-churn" min="0" max="25" step="0.5" value="5.0" oninput="updateModel()"/>
  </div>
  <div class="control-group">
    <div class="control-label"><span>Fixed OpEx / Mo</span><span class="control-val" id="val-fixed">$5,000</span></div>
    <input type="range" id="param-fixed" min="500" max="25000" step="250" value="5000" oninput="updateModel()"/>
  </div>
  <div class="control-group">
    <div class="control-label"><span>COGS per Client</span><span class="control-val" id="val-cogs">$30</span></div>
    <input type="range" id="param-cogs" min="0" max="300" step="5" value="30" oninput="updateModel()"/>
  </div>
</div>

<div>
  <div class="kpi-row">
    <div class="kpi-card"><span class="kpi-label">Break-Even Volume</span><div class="kpi-val" id="kpi-breakeven-clients">23</div><span id="kpi-breakeven-rev" style="font-size:12px;color:var(--text-muted);">$5,750 / mo threshold</span></div>
    <div class="kpi-card"><span class="kpi-label">Unit Margin</span><div class="kpi-val good" id="kpi-unit-margin">$220</div><span id="kpi-margin-pct" style="font-size:12px;color:var(--text-muted);">88.0% margin</span></div>
    <div class="kpi-card"><span class="kpi-label">Month 1 Profit</span><div class="kpi-val good" id="kpi-m1-profit">$2,260</div><span id="kpi-m1-rev" style="font-size:12px;color:var(--text-muted);">From $8,250 rev</span></div>
    <div class="kpi-card"><span class="kpi-label">Month 12 Run-Rate</span><div class="kpi-val rev" id="kpi-m12-profit">$8,420</div><span id="kpi-m12-clients" style="font-size:12px;color:var(--text-muted);">61 active clients</span></div>
  </div>
  <div class="chart-box"><div style="margin-bottom:12px;font-weight:700;color:var(--text-muted);">12-MONTH PROJECTIONS (REVENUE VS EXPENSES VS PROFIT)</div><div style="flex-grow:1;position:relative;"><canvas id="scenarioChart"></canvas></div>
</div>
</div>

<footer>
  <div><a href="https://3sci.com" target="_blank" class="footer-link">3sci.com</a> • Open Source Tools for Founders</div>
  <span>100% Client-Side • Zero Telemetry</span>
</footer>

<script>
  let chartInstance = null;
  const PRESETS = {
    conservative: { price: 200, clients: 20, newClients: 2, churn: 8.0, fixed: 4500, cogs: 35 },
    base: { price: 250, clients: 30, newClients: 5, churn: 5.0, fixed: 5000, cogs: 30 },
    aggressive: { price: 300, clients: 35, newClients: 9, churn: 3.5, fixed: 6500, cogs: 25 }
  };
  function setPreset(key) {
    const p = PRESETS[key];
    ['price','clients','newClients','churn','fixed','cogs'].forEach(k => {
      const id = k === 'newClients' ? 'param-new-clients' : 'param-' + k;

```

```

    document.getElementById(id).value = p[k];
  });
  updateModel();
}
function updateModel() {
  const price = parseFloat(document.getElementById('param-price').value);
  const starting = parseInt(document.getElementById('param-clients').value);
  const added = parseInt(document.getElementById('param-new-clients').value);
  const churn = parseFloat(document.getElementById('param-churn').value) / 100;
  const fixed = parseFloat(document.getElementById('param-fixed').value);
  const cogs = parseFloat(document.getElementById('param-cogs').value);

  document.getElementById('val-price').innerText = '$' + price.toLocaleString();
  document.getElementById('val-clients').innerText = starting;
  document.getElementById('val-new-clients').innerText = added;
  document.getElementById('val-churn').innerText = (churn * 100).toFixed(1) + '%';
  document.getElementById('val-fixed').innerText = '$' + fixed.toLocaleString();
  document.getElementById('val-cogs').innerText = '$' + cogs.toLocaleString();

  const unitMargin = price - cogs;
  const marginPct = (unitMargin / price) * 100;
  const beClients = unitMargin > 0 ? Math.ceil(fixed / unitMargin) : '∞';

  document.getElementById('kpi-breakeven-clients').innerText = beClients;
  document.getElementById('kpi-breakeven-rev').innerText = unitMargin > 0 ? '$' + (beClients * price).toLocaleString()
+ ' / mo' : 'Unprofitable';
  document.getElementById('kpi-unit-margin').innerText = '$' + unitMargin.toLocaleString();
  document.getElementById('kpi-margin-pct').innerText = marginPct.toFixed(1) + '% contribution';

  let current = starting, labels = [], revData = [], costData = [], profitData = [];
  for (let m = 1; m <= 12; m++) {
    labels.push('M' + m);
    current = Math.max(0, current - Math.round(current * churn) + added);
    const rev = current * price, cost = fixed + (current * cogs), profit = rev - cost;
    revData.push(rev); costData.push(cost); profitData.push(profit);
    if (m === 1) {
      document.getElementById('kpi-m1-profit').innerText = '$' + profit.toLocaleString();
      document.getElementById('kpi-m1-rev').innerText = 'From $' + rev.toLocaleString() + ' rev';
    }
    if (m === 12) {
      document.getElementById('kpi-m12-profit').innerText = '$' + profit.toLocaleString();
      document.getElementById('kpi-m12-clients').innerText = current + ' clients ($' + rev.toLocaleString() + ' rev)';
    }
  }
  renderChart(labels, revData, costData, profitData);
}
function renderChart(labels, revData, costData, profitData) {
  const ctx = document.getElementById('scenarioChart').getContext('2d');
  if (chartInstance) chartInstance.destroy();
  chartInstance = new Chart(ctx, {
    type: 'line',
    data: {
      labels: labels,
      datasets: [
        { label: 'Gross Revenue', data: revData, borderColor: '#38bdf8', borderWidth: 3, tension: 0.3, fill: false },
        { label: 'Total Costs', data: costData, borderColor: '#ef4444', borderDash: [5,5], borderWidth: 2, tension: 0.3, fill: false },
        { label: 'Net Profit', data: profitData, borderColor: '#10b981', backgroundColor: 'rgba(16,185,129,0.15)', borderWidth: 3, tension: 0.3, fill: true }
      ]
    },
    options: {
      responsive: true, maintainAspectRatio: false,
      plugins: { legend: { labels: { color: '#cbd5e1' } } },
      scales: {
        x: { grid: { color: '#1e2c3d' }, ticks: { color: '#94a3b8' } },
        y: { grid: { color: '#1e2c3d' }, ticks: { color: '#94a3b8', callback: v => '$' + v.toLocaleString() } }
      }
    }
  });
}
}

```

```
    window.onload = updateModel;  
</script>  
</body>  
</html>
```

4. How to Customize Parameters for Different Business Models

You can adapt this tool for consulting agencies, SaaS subscriptions, boutique coaching, or e-commerce memberships without writing custom software.

RULE 1

Changing Slider Minimums & Maximums

Open `calculator.html` in Notepad and find the slider inputs. For example, if you sell high-ticket retainers instead of small subscriptions, change the price slider:

```
<input type="range" id="param-price" min="1000" max="10000" step="250" value="3500" oninput="updateModel()">
```

This expands your slider to evaluate retainers between \$1,000 and \$10,000 in \$250 steps.

RULE 2

Modifying Default Presets

In the script tag, locate the `PRESETS` object. You can replace the default scenarios with your firm's actual historical numbers:

```
const PRESETS = {  
  conservative: { price: 3000, clients: 5, newClients: 1, churn: 5.0, fixed: 8000, cogs: 400 },  
  base: { price: 4500, clients: 8, newClients: 2, churn: 3.0, fixed: 10000, cogs: 350 },  
  aggressive: { price: 6000, clients: 12, newClients: 4, churn: 2.0, fixed: 14000, cogs: 300 }  
};
```

🌟 Always ask your preferred AI platform if you need help updating code!

You don't need to be a web developer to customize this simulator. If you want to add employee capacity limits, convert the currency to Euros or Pounds, or change the color palette, simply paste the code of `calculator.html` into ChatGPT, Claude, or Gemini with this prompt:

"Here is my calculator.html file. Please preserve all existing functionality and Chart.js logic, but update the currency to [EUR/GBP] and add an extra slider for [Your New Parameter]."

Your AI platform will provide the complete, updated code block ready for you to copy and paste.

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